Arithmetic Sequences as Linear Functions

Writing recursive rules from listed sequences and from real contexts, running a recursive rule forward, and converting a recursive rule to an explicit linear function

Directions. A recursive formula for an arithmetic sequence always has **two** lines — the starting value and the step:

$$a_1 = \text{first term}, \qquad a_n = a_{n-1} + d \quad (n \geq 2).$$

- Write *both* lines for every problem, then answer the question asked. A rule with no starting value names infinitely many sequences.
- Keep the units of the situation in your answer where the problem states them.
- Getting from a_1 to a_n takes $n - 1$ steps, not n.

1. **From a list.** A tiling pattern uses this many tiles in each stage:

$$5, \quad 12, \quad 19, \quad 26, \quad \ldots$$

Write the recursive formula for a_n, then find a_6, the number of tiles in stage 6.

Answer: ________________

2. From a list, decreasing. A drone descends in equal steps, and its altitude in meters at each reading is:

$$48, \quad 41, \quad 34, \quad 27, \quad \ldots$$

Write the recursive formula for a_n, then find a_9. Say what a negative value of a_9 means for the drone.

Answer: ________________

3. Stacked chairs. One chair on the floor stands 82 centimeters tall. Each extra chair stacked on top adds 9 centimeters to the height of the stack. Let a_n be the height in centimeters of a stack of n chairs.

Write the recursive formula for a_n, stating a_1 and the step, then find the height of a stack of 12 chairs.

Answer: ________________ cm

4. Run the rule forward. A sequence is defined by

$$a_1 = 7, \qquad a_n = a_{n-1} + 6 \quad (n \geq 2).$$

Find a_{15} without listing every term, and explain in one sentence how many times the step of 6 is applied.

Answer: ________________

5. Run the rule backward. A sequence is defined by

$$a_1 = 100, \qquad a_n = a_{n-1} - 12 \quad (n \geq 2).$$

Which term of this sequence equals 16? Set up an equation in n and solve it; do not answer by listing terms.

Answer: ________________

6. Draining tank. A tank holds 240 liters of water. At the end of every hour a pump removes 15 liters. Let a_n be the number of liters left at the end of hour n.

Write the recursive formula for a_n, being careful about what a_1 is, then find how many liters are left at the end of hour 8.

Answer: ________________ L

7. Gym membership. A gym charges a one-time joining fee of 60 dollars, then 35 dollars at the end of each month. Let a_n be the total amount paid, in dollars, after n months of membership.

Write the recursive formula for a_n, convert it to an explicit rule of the form $a_n = mn + b$, and use the explicit rule to find the total paid after 24 months.

Answer: ________________

8. Recursive to explicit. A sequence is defined by

$$a_1 = -5, \qquad a_n = a_{n-1} + 4 \quad (n \geq 2).$$

Write the explicit rule as a linear function of n, name the slope and the value of the function at $n = 0$, and find a_{20}.

Answer: __________________

9. Missing first term. A table of an arithmetic sequence records only two of its terms.

n	3	7
a_n	23	47

Find the common difference, then work back to a_1 and write the full recursive formula. Give the common difference and a_1 as your two answers.

Answer: ______________

10. Synthesis: theater seating. Row 1 of a theater has 18 seats, and each row behind it has 3 more seats than the row in front of it.

Write the recursive formula for a_n, the number of seats in row n. Then find the **total** number of seats in the first 15 rows, and explain why adding the first and last row and multiplying by half the number of rows gives the same total.

Answer: __________________